

## DAUGAVPILS, Latvia

WMO: 265440

Lat: 55.8697N

Lon: 26.6175E

Elev: 325

StdP: 14.52

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -8.0       | -0.8 | -12.7                         | 2.8 | -7.1 | -5.6 | 4.1 | 0.4  | 19.8                  | 30.6 | 18.0 | 30.4 | 2.4                   | 160  | 0.552 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.9                   | 82.5            | 67.2 | 79.4 | 65.3 | 76.4 | 63.9 | 69.7                | 78.6 | 67.6 | 75.8 | 65.8 | 73.1 | 6.5                  | 150  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.5                            | 98.8 | 74.4 | 64.7 | 92.5 | 71.6 | 63.0 | 87.1 | 69.7 | 33.9          | 78.7 | 32.2 | 75.8 | 30.8 | 73.2 | 79.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 16.2          | 14.2 | 12.6 | DB                         | -15.3 | 87.7               | 8.2 | 3.2                                                | -21.2 | 90.0       | -25.9 | 91.8       | -30.5 | 93.6       | -36.4 | 95.9 | (4) |
| (5)               |      |      | WB                         | -15.7 | 72.4               | 7.9 | 2.1                                                | -21.3 | 73.9       | -25.9 | 75.1       | -30.4 | 76.3       | -36.1 | 77.8 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 43.6   | 23.2  | 23.6  | 31.4  | 44.0  | 53.6  | 59.9  | 64.3  | 62.0  | 53.5  | 43.5  | 34.6  | 27.7  |  |  |
|                                                                        | DBStd       | 16.71  | 11.61 | 11.90 | 8.50  | 7.91  | 7.93  | 6.12  | 5.48  | 5.57  | 6.24  | 7.08  | 8.26  | 10.10 |  |  |
|                                                                        | HDD50       | 3824   | 830   | 740   | 576   | 212   | 51    | 2     | 0     | 0     | 36    | 223   | 461   | 693   |  |  |
|                                                                        | HDD65       | 7956   | 1295  | 1160  | 1040  | 630   | 362   | 175   | 81    | 129   | 348   | 667   | 911   | 1158  |  |  |
|                                                                        | CDD50       | 1474   | 0     | 0     | 0     | 33    | 164   | 301   | 443   | 372   | 140   | 21    | 0     | 0     |  |  |
|                                                                        | CDD65       | 128    | 0     | 0     | 0     | 0     | 9     | 23    | 59    | 35    | 2     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 1186   | 0     | 0     | 0     | 5     | 92    | 208   | 539   | 323   | 19    | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 221    | 0     | 0     | 0     | 0     | 9     | 27    | 121   | 63    | 1     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 5.8    | 6.7   | 6.4   | 6.6   | 6.0   | 5.7   | 5.3   | 4.6   | 4.6   | 5.0   | 6.0   | 6.6   | 6.5   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 24.8   | 1.4   | 1.1   | 1.3   | 1.5   | 2.0   | 2.8   | 3.1   | 2.7   | 2.6   | 2.0   | 2.1   | 1.8   |  |  |
|                                                                        | PrecMax     | 35.8   | 3.2   | 2.0   | 3.2   | 4.9   | 4.5   | 5.8   | 8.0   | 6.4   | 5.9   | 5.0   | 5.1   | 3.3   |  |  |
|                                                                        | PrecMin     | 17.9   | 0.3   | 0.1   | 0.2   | 0.1   | 0.5   | 0.7   | 0.2   | 0.0   | 0.0   | 0.3   | 0.0   | 0.3   |  |  |
|                                                                        | PrecStd     | 4.0    | 0.7   | 0.5   | 0.7   | 1.0   | 0.9   | 1.2   | 1.8   | 1.5   | 1.2   | 1.1   | 0.9   | 0.6   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 44.6  | 43.9  | 56.9  | 74.0  | 81.4  | 83.3  | 88.2  | 85.9  | 76.8  | 64.4  | 51.8  | 47.6  |  |  |
|                                                                        |             | MCWB   | 42.9  | 40.4  | 45.1  | 56.2  | 64.6  | 66.6  | 69.2  | 71.3  | 63.0  | 57.3  | 49.1  | 45.6  |  |  |
|                                                                        | 2%          | DB     | 40.3  | 41.0  | 50.6  | 68.1  | 76.4  | 79.4  | 83.5  | 81.1  | 71.4  | 59.2  | 48.9  | 44.4  |  |  |
|                                                                        |             | MCWB   | 38.8  | 38.3  | 42.2  | 53.1  | 62.5  | 64.9  | 68.4  | 66.4  | 61.4  | 54.3  | 46.7  | 42.6  |  |  |
|                                                                        | 5%          | DB     | 37.7  | 38.7  | 46.0  | 63.1  | 72.6  | 76.1  | 79.9  | 77.5  | 67.5  | 56.3  | 46.7  | 41.6  |  |  |
|                                                                        |             | MCWB   | 36.3  | 36.7  | 40.4  | 50.4  | 59.3  | 63.3  | 66.4  | 64.3  | 59.3  | 52.6  | 44.9  | 40.2  |  |  |
|                                                                        | 10%         | DB     | 35.7  | 36.8  | 42.4  | 58.6  | 68.8  | 72.6  | 76.5  | 73.8  | 64.3  | 53.8  | 44.8  | 38.9  |  |  |
|                                                                        |             | MCWB   | 34.5  | 35.3  | 38.5  | 48.2  | 57.1  | 61.7  | 65.1  | 63.2  | 57.6  | 50.9  | 43.3  | 37.5  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 43.0  | 41.3  | 46.8  | 57.9  | 67.1  | 70.1  | 72.6  | 72.3  | 65.6  | 58.5  | 49.8  | 46.3  |  |  |
|                                                                        |             | MCDB   | 44.3  | 43.3  | 54.4  | 70.1  | 77.7  | 79.6  | 83.2  | 82.8  | 71.8  | 63.0  | 51.2  | 47.3  |  |  |
|                                                                        | 2%          | WB     | 39.0  | 38.7  | 43.5  | 54.4  | 63.9  | 67.2  | 70.4  | 69.0  | 62.6  | 55.2  | 47.0  | 42.6  |  |  |
|                                                                        |             | MCDB   | 40.0  | 40.2  | 48.5  | 65.1  | 73.5  | 75.8  | 79.5  | 77.8  | 69.2  | 58.0  | 48.3  | 43.7  |  |  |
|                                                                        | 5%          | WB     | 36.6  | 36.9  | 41.2  | 51.9  | 61.2  | 65.1  | 68.3  | 66.6  | 60.5  | 53.1  | 45.1  | 40.3  |  |  |
|                                                                        |             | MCDB   | 37.5  | 38.3  | 44.9  | 61.0  | 70.2  | 72.8  | 76.5  | 73.7  | 65.9  | 55.8  | 46.4  | 41.7  |  |  |
|                                                                        | 10%         | WB     | 34.7  | 35.4  | 38.9  | 49.4  | 58.5  | 63.1  | 66.4  | 64.5  | 58.4  | 51.2  | 43.4  | 37.7  |  |  |
|                                                                        |             | MCDB   | 35.6  | 36.8  | 42.1  | 56.8  | 66.6  | 70.2  | 73.6  | 71.0  | 62.9  | 53.7  | 44.6  | 38.9  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 8.3   | 10.4  | 13.0  | 18.2  | 19.6  | 17.7  | 17.9  | 18.3  | 16.0  | 11.3  | 7.1   | 6.8   |  |  |
|                                                                        |             | MCDBR  | 6.5   | 7.6   | 17.9  | 26.9  | 24.5  | 23.0  | 23.8  | 24.1  | 21.3  | 14.6  | 8.7   | 6.6   |  |  |
|                                                                        | 5% WB       | MCWBR  | 6.4   | 6.3   | 11.3  | 13.6  | 11.8  | 10.6  | 10.4  | 10.8  | 11.8  | 9.9   | 7.6   | 6.3   |  |  |
|                                                                        |             | MCWBR  | 6.5   | 7.2   | 14.9  | 22.9  | 21.6  | 18.7  | 19.6  | 20.0  | 18.2  | 13.0  | 8.1   | 6.4   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.289 | 0.323 | 0.356 | 0.398 | 0.389 | 0.379 | 0.412 | 0.411 | 0.374 | 0.350 | 0.331 | 0.290 |  |  |
|                                                                        | taud        |        | 2.175 | 2.148 | 2.186 | 2.223 | 2.320 | 2.373 | 2.303 | 2.296 | 2.373 | 2.393 | 2.257 | 2.193 |  |  |
|                                                                        | Ebn at Noon |        | 191   | 229   | 252   | 259   | 270   | 274   | 261   | 253   | 247   | 222   | 174   | 160   |  |  |
|                                                                        | Edh at Noon |        | 18    | 28    | 34    | 39    | 37    | 36    | 38    | 36    | 29    | 22    | 17    | 15    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 168   | 426   | 863   | 1270  | 1676  | 1768  | 1653  | 1362  | 904   | 425   | 141   | 101   |  |  |
|                                                                        | RadStd      |        | 40    | 82    | 134   | 150   | 165   | 143   | 168   | 122   | 85    | 79    | 24    | 23    |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (18 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page